

#Next Word Predictor

```

faqs = """About the Program
What is the course fee for Data Science Mentorship Program (DSMP 2023)
The course follows a monthly subscription model where you have to make monthly payments of Rs 799/month.
What is the total duration of the course?
The total duration of the course is 7 months. So the total course fee becomes 799*7 = Rs 5600(approx.)
What is the syllabus of the mentorship program?
We will be covering the following modules:
Python Fundamentals
Python libraries for Data Science
Data Analysis
SQL for Data Science
Maths for Machine Learning
ML Algorithms
Practical ML
MLOPs
Case studies
You can check the detailed syllabus here - https://learnwith.campusx.in/courses/CampusX-Data-Science-Mentorship-Program-637339af
Will Deep Learning and NLP be a part of this program?
No, NLP and Deep Learning both are not a part of this program's curriculum.
What if I miss a live session? Will I get a recording of the session?
Yes all our sessions are recorded, so even if you miss a session you can go back and watch the recording.
Where can I find the class schedule?
Checkout this google sheet to see month by month time table of the course - https://docs.google.com/spreadsheets/d/160oTax\_A60RA
What is the time duration of all the live sessions?
Roughly, all the sessions last 2 hours.
What is the language spoken by the instructor during the sessions?
Hinglish
How will I be informed about the upcoming class?
You will get a mail from our side before every paid session once you become a paid user.
Can I do this course if I am from a non-tech background?
Yes, absolutely.
I am late, can I join the program in the middle?
Absolutely, you can join the program anytime.
If I join/pay in the middle, will I be able to see all the past lectures?
Yes, once you make the payment you will be able to see all the past content in your dashboard.
Where do I have to submit the task?
You don't have to submit the task. We will provide you with the solutions, you have to self evaluate the task yourself.
Will we do case studies in the program?
Yes.
Where can we contact you?
You can mail us at nitish.campusx@gmail.com
Payment/Registration related questions
Where do we have to make our payments? Your YouTube channel or website?
You have to make all your monthly payments on our website. Here is the link for our website - https://learnwith.campusx.in/
Can we pay the entire amount of Rs 5600 all at once?
Unfortunately no, the program follows a monthly subscription model.
What is the validity of monthly subscription? Suppose if I pay on 15th Jan, then do I have to pay again on 1st Feb or 15th Feb
15th Feb. The validity period is 30 days from the day you make the payment. So essentially you can join anytime you don't have to
What if I don't like the course after making the payment. What is the refund policy?
You get a 7 days refund period from the day you have made the payment.
I am living outside India and I am not able to make the payment on the website, what should I do?
You have to contact us by sending a mail at nitish.campusx@gmail.com
Post registration queries
Till when can I view the paid videos on the website?
This one is tricky, so read carefully. You can watch the videos till your subscription is valid. Suppose you have purchased subscription
But once the course is over and you have paid us Rs 5600(or 7 installments of Rs 799) you will be able to watch the paid sessions.
Why lifetime validity is not provided?
Because of the low course fee.
Where can I reach out in case of a doubt after the session?
You will have to fill a google form provided in your dashboard and our team will contact you for a 1 on 1 doubt clearance session.
If I join the program late, can I still ask past week doubts?
Yes, just select past week doubt in the doubt clearance google form.
I am living outside India and I am not able to make the payment on the website, what should I do?
You have to contact us by sending a mail at nitish.campusx@gmail.com
Certificate and Placement Assistance related queries
What is the criteria to get the certificate?
There are 2 criterias:
You have to pay the entire fee of Rs 5600
You have to attempt all the course assessments.
I am joining late. How can I pay payment of the earlier months?
You will get a link to pay fee of earlier months in your dashboard once you pay for the current month.
I have read that Placement assistance is a part of this program. What comes under Placement assistance?
This is to clarify that Placement assistance does not mean Placement guarantee. So we don't guarantee you any jobs or for that matter

```

```
Portfolio Building sessions
Soft skill sessions
Sessions with industry mentors
Discussion on Job hunting strategies
"""
```

```
#assigning numbers(index) to each word

#importing important libraries

import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
```

```
tokenizer = Tokenizer()
```

```
tokenizer.fit_on_texts([faqs])
```

```
tokenizer.word_index
```

```
{
  'side': 196,
  'before': 197,
  'every': 198,
  'become': 199,
  'user': 200,
  'non': 201,
  'tech': 202,
  'background': 203,
  'lectures': 204,
  'content': 205,
  'provide': 206,
  'solutions': 207,
  'self': 208,
  'evaluate': 209,
  'yourself': 210,
  'questions': 211,
  'youtube': 212,
  'channel': 213,
  'amount': 214,
  'unfortunately': 215,
  'then': 216,
  '1st': 217,
  '30': 218,
  'essentially': 219,
  'wait': 220,
  'end': 221,
  'like': 222,
  'making': 223,
  'policy': 224,
  'made': 225,
  'post': 226,
  'when': 227,
  'view': 228,
  'one': 229,
  'tricky': 230,
  'carefully': 231,
  'valid': 232,
  'purchased': 233,
  '20th': 234,
  'purchase': 235,
  'over': 236,
  'installments': 237,
  'aug': 238,
  '2024': 239,
  'why': 240,
  'lifetime': 241,
  'because': 242,
  'low': 243,
  'reach': 244,
  'out': 245,
  'fill': 246,
  'team': 247,
  'still': 248,
  'ask': 249,
  'doubts': 250,
  'select': 251,
  'gmai': 252,
  'criteria': 253,
}
```

```
#find out all the sentences present in the dataset
```

```
for sentence in faqs.split('\n'):
    print(sentence)
```

About the Program

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The total duration of the course is 7 months. So the total course fee becomes $799 \times 7 = \text{Rs } 5600$ (approx.)

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Practical ML

MLOPs

Case studies

You can check the detailed syllabus here - <https://learnwith.campusx.in/courses/CampusX-Data-Science-Mentorship-Program-637339>

Will Deep Learning and NLP be a part of this program?

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What if I miss a live session? Will I get a recording of the session?

Yes all our sessions are recorded, so even if you miss a session you can go back and watch the recording.

Where can I find the class schedule?

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Hinglish

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Why lifetime validity is not provided?

~ ~ ~ ~ ~

```
for sentence in faqs.split('\n'):
    print(tokenizer.texts_to_sequences([sentence])[0])# added 0 to bring it in a single list
```

```
[93, 1, 13]
[11, 7, 1, 12, 42, 15, 43, 53, 68, 13, 147, 148]
[1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44, 69, 6, 31, 70, 45]
[11, 7, 1, 71, 72, 6, 1, 12]
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149, 70, 54, 31, 55, 150]
[11, 7, 1, 96, 6, 1, 68, 13]
[27, 9, 19, 151, 1, 152, 153]
[97, 154]
[97, 155, 15, 43, 53]
[43, 156]
[157, 15, 43, 53]
[158, 15, 159, 74]
[98, 160]
[161, 98]
[162]
```

```

[75, 99]
[2, 10, 163, 1, 164, 96, 76, 77, 100, 33, 14, 165, 33, 43, 53, 68, 13, 166]
[9, 101, 74, 20, 102, 19, 5, 78, 6, 21, 13]
[103, 102, 20, 101, 74, 167, 56, 46, 5, 78, 6, 21, 168, 169]
[11, 22, 3, 104, 5, 105, 34, 9, 3, 47, 5, 106, 6, 1, 34]
[48, 16, 35, 17, 56, 170, 32, 107, 22, 2, 104, 5, 34, 2, 10, 171, 172, 20, 57, 1, 106]
[30, 10, 3, 173, 1, 108, 174]
[175, 21, 58, 176, 4, 79, 45, 59, 45, 109, 177, 6, 1, 12, 77, 178, 58, 60, 179, 180, 181, 182, 183, 184, 185]
[11, 7, 1, 109, 72, 6, 16, 1, 105, 17]
[186, 16, 1, 17, 187, 110, 188]
[11, 7, 1, 189, 190, 59, 1, 191, 192, 1, 17]
[193]
[111, 9, 3, 19, 194, 93, 1, 195, 108]
[2, 9, 47, 5, 61, 62, 35, 196, 197, 198, 36, 34, 49, 2, 199, 5, 36, 200]
[10, 3, 28, 21, 12, 22, 3, 23, 62, 5, 201, 202, 203]
[48, 112]
[3, 23, 80, 10, 3, 37, 1, 13, 14, 1, 113]
[112, 2, 10, 37, 1, 13, 114]
[22, 3, 37, 24, 14, 1, 113, 9, 3, 19, 38, 4, 79, 16, 1, 50, 204]
[48, 49, 2, 26, 1, 25, 2, 9, 19, 38, 4, 79, 16, 1, 50, 205, 14, 39, 81]
[30, 28, 3, 8, 4, 115, 1, 82]
[2, 83, 8, 4, 115, 1, 82, 27, 9, 206, 2, 116, 1, 207, 2, 8, 4, 208, 209, 1, 82, 210]
[9, 27, 28, 75, 99, 14, 1, 13]
[48]
[30, 10, 27, 63, 2]
[2, 10, 61, 64, 65, 84, 33, 117, 60]
[25, 118, 119, 211]
[30, 28, 27, 8, 4, 26, 35, 69, 39, 212, 213, 66, 40]
[2, 8, 4, 26, 16, 39, 44, 69, 18, 35, 40, 76, 7, 1, 120, 15, 35, 40, 77, 100, 33, 14]
[10, 27, 24, 1, 121, 214, 6, 31, 55, 16, 65, 49]
[215, 103, 1, 13, 94, 5, 44, 29, 95]
[11, 7, 1, 85, 6, 44, 29, 122, 22, 3, 24, 18, 86, 87, 216, 28, 3, 8, 4, 24, 123, 18, 217, 51, 66, 86, 51]
[86, 51, 1, 85, 88, 7, 218, 124, 62, 1, 125, 2, 26, 1, 25, 32, 219, 2, 10, 37, 114, 2, 83, 8, 4, 220, 15, 5, 45, 4, 221]
[11, 22, 3, 83, 222, 1, 12, 89, 223, 1, 25, 11, 7, 1, 126, 224]
[2, 47, 5, 54, 124, 126, 88, 62, 1, 125, 2, 8, 225, 1, 25]
[3, 23, 127, 128, 129, 20, 3, 23, 46, 38, 4, 26, 1, 25, 18, 1, 40, 11, 130, 3, 28]
[2, 8, 4, 63, 64, 59, 131, 5, 61, 65, 84, 33, 117, 60]
[226, 118, 132]
[90, 227, 10, 3, 228, 1, 36, 133, 18, 1, 40]
[21, 229, 7, 230, 32, 134, 231, 2, 10, 57, 1, 133, 90, 39, 29, 7, 232, 122, 2, 8, 233, 29, 18, 91, 87, 2, 9, 19, 38, 4, 57, 16]
[135, 49, 1, 12, 7, 236, 20, 2, 8, 36, 64, 31, 55, 66, 54, 237, 6, 31, 70, 2, 9, 19, 38, 4, 57, 1, 36, 17, 90, 238, 239]
[240, 241, 85, 7, 46, 136]

```

```

#creating dataset from the above list
input_sequences=[]
for sentence in faqs.split('\n'):
    tokenized_sentence= tokenizer.texts_to_sequences([sentence])[0]

    for i in range(1,len(tokenized_sentence)):
        input_sequences.append(tokenized_sentence[:i+1])

```

input_sequences

```

[[93, 1],
 [93, 1, 13],
 [11, 7],
 [11, 7, 1],
 [11, 7, 1, 12],
 [11, 7, 1, 12, 42],
 [11, 7, 1, 12, 42, 15],
 [11, 7, 1, 12, 42, 15, 43],
 [11, 7, 1, 12, 42, 15, 43, 53],
 [11, 7, 1, 12, 42, 15, 43, 53, 68],
 [11, 7, 1, 12, 42, 15, 43, 53, 68, 13],
 [11, 7, 1, 12, 42, 15, 43, 53, 68, 13, 147],
 [11, 7, 1, 12, 42, 15, 43, 53, 68, 13, 147, 148],
 [1, 12],
 [1, 12, 94],
 [1, 12, 94, 5],
 [1, 12, 94, 5, 44],
 [1, 12, 94, 5, 44, 29],
 [1, 12, 94, 5, 44, 29, 95],
 [1, 12, 94, 5, 44, 29, 95, 30],
 [1, 12, 94, 5, 44, 29, 95, 30, 2],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44, 69],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44, 69, 6],
 [1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44, 69, 6, 31],

```

```
[1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44, 69, 6, 31, 70],
[1, 12, 94, 5, 44, 29, 95, 30, 2, 8, 4, 26, 44, 69, 6, 31, 70, 45],
[11, 7],
[11, 7, 1],
[11, 7, 1, 71],
[11, 7, 1, 71, 72],
[11, 7, 1, 71, 72, 6],
[11, 7, 1, 71, 72, 6, 1],
[11, 7, 1, 71, 72, 6, 1, 12],
[1, 71],
[1, 71, 72],
[1, 71, 72, 6],
[1, 71, 72, 6, 1],
[1, 71, 72, 6, 1, 12],
[1, 71, 72, 6, 1, 12, 7],
[1, 71, 72, 6, 1, 12, 7, 54],
[1, 71, 72, 6, 1, 12, 7, 54, 73],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149, 70],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149, 70, 54],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149, 70, 54, 31],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149, 70, 54, 31, 55],
[1, 71, 72, 6, 1, 12, 7, 54, 73, 32, 1, 71, 12, 42, 149, 70, 54, 31, 55, 150],
[11, 7],
[11, 7, 1]
```

```
#for same input size, we'll add zero padding
max_len = max([len(x) for x in input_sequences])
```

```
from tensorflow.keras.preprocessing.sequence import pad_sequences
padded_input_sequences= pad_sequences(input_sequences,maxlen=max_len,padding='pre')
```

padded_input_sequences

```
array([[ 0,  0,  0, ...,  0, 93,  1],
       [ 0,  0,  0, ..., 93,  1, 13],
       [ 0,  0,  0, ...,  0, 11,  7],
       ...,
       [ 0,  0,  0, ..., 279, 18, 280],
       [ 0,  0,  0, ..., 18, 280, 281],
       [ 0,  0,  0, ..., 280, 281, 282]], dtype=int32)
```

```
X= padded_input_sequences[:, :-1] #all rows and upto second last column
```

padded_input_sequences[:, :-1]

```
array([[ 1, 13,  7,  1, 12, 42, 15, 43, 53, 68, 13, 147, 148,
        12, 94,  5, 44, 29, 95, 30,  2,  8,  4, 26, 44, 69,
         6, 31, 70, 45,  7,  1, 71, 72,  6,  1, 12, 71, 72,
         6,  1, 12,  7, 54, 73, 32,  1, 71, 12, 42, 149, 70,
        54, 31, 55, 150,  7,  1, 96,  6,  1, 68, 13,  9, 19,
       151,  1, 152, 153, 154, 155, 15, 43, 53, 156, 15, 43, 53,
       15, 159, 74, 160, 98, 99, 10, 163,  1, 164, 96, 76, 77,
      100, 33, 14, 165, 33, 43, 53, 68, 13, 166, 101, 74, 20,
      102, 19,  5, 78,  6, 21, 13, 102, 20, 101, 74, 167, 56,
       46,  5, 78,  6, 21, 168, 169, 22,  3, 104,  5, 105, 34,
         9,  3, 47,  5, 106,  6,  1, 34, 16, 35, 17, 56, 170,
       32, 107, 22,  2, 104,  5, 34,  2, 10, 171, 172, 20, 57,
         1, 106, 10,  3, 173,  1, 108, 174, 21, 58, 176,  4, 79,
       45, 59, 45, 109, 177,  6,  1, 12, 77, 178, 58, 60, 179,
      180, 181, 182, 183, 184, 185,  7,  1, 109, 72,  6, 16,  1,
      105, 17, 16,  1, 17, 187, 110, 188,  7,  1, 189, 190, 59,
         1, 191, 192,  1, 17,  9,  3, 19, 194, 93,  1, 195, 108,
         9, 47,  5, 61, 62, 35, 196, 197, 198, 36, 34, 49,  2,
      199,  5, 36, 200,  3, 28, 21, 12, 22,  3, 23, 62,  5,
      201, 202, 203, 112, 23, 80, 10,  3, 37,  1, 13, 14,  1,
      113,  2, 10, 37,  1, 13, 114,  3, 37, 24, 14,  1, 113,
         9,  3, 19, 38,  4, 79, 16,  1, 50, 204, 49,  2, 26,
         1, 25,  2,  9, 19, 38,  4, 79, 16,  1, 50, 205, 14,
        39, 81, 28,  3,  8,  4, 115,  1, 82, 83,  8,  4, 115,
         1, 82, 27,  9, 206,  2, 116,  1, 207,  2,  8,  4, 208,
      209,  1, 82, 210, 27, 28, 75, 99, 14,  1, 13, 10, 27,
        63,  2, 10, 61, 64, 65, 84, 33, 117, 60, 118, 119, 211,
        28, 27,  8,  4, 26, 35, 69, 39, 212, 213, 66, 40,  8,
         4, 26, 16, 39, 44, 69, 18, 35, 40, 76,  7,  1, 120,
```

```

15, 35, 40, 77, 100, 33, 14, 27, 24, 1, 121, 214, 6,
31, 55, 16, 65, 49, 103, 1, 13, 94, 5, 44, 29, 95,
7, 1, 85, 6, 44, 29, 122, 22, 3, 24, 18, 86, 87,
216, 28, 3, 8, 4, 24, 123, 18, 217, 51, 66, 86, 51,
51, 1, 85, 88, 7, 218, 124, 62, 1, 125, 2, 26, 1,
25, 32, 219, 2, 10, 37, 114, 2, 83, 8, 4, 220, 15,
5, 45, 4, 221, 22, 3, 83, 222, 1, 12, 89, 223, 1,
25, 11, 7, 1, 126, 224, 47, 5, 54, 124, 126, 88, 62,
1, 125, 2, 8, 225, 1, 25, 23, 127, 128, 129, 20, 3,
23, 46, 38, 4, 26, 1, 25, 18, 1, 40, 11, 130, 3,
28, 8, 4, 63, 64, 59, 131, 5, 61, 65, 84, 33, 117,
60, 118, 132, 227, 10, 3, 228, 1, 36, 133, 18, 1, 40,
229, 7, 230, 32, 134, 231, 2, 10, 57, 1, 133, 90, 39,
29, 7, 232, 122, 2, 8, 233, 29, 18, 91, 87, 2, 9,
19, 38, 4, 57, 16, 1, 50, 36, 17, 14, 1, 88, 6,
91, 87, 4, 234, 51, 135, 89, 91, 51, 2, 9, 8, 4,
235, 1, 29, 123, 49, 1, 12, 7, 236, 20, 2, 8, 36,
64, 31, 55, 66, 54, 237, 6, 31, 70, 2, 9, 19, 38,
4, 57, 1, 36, 17, 90, 238, 239, 241, 85, 7, 46, 136,
6, 1, 243, 12, 42, 10, 3, 244, 245, 14, 75, 6, 5,
67, 89, 1, 34, 9, 8, 4, 246, 5, 58, 137, 136, 14,
39, 81, 20, 35, 247, 9, 63, 2, 15, 5, 138, 18, 138,
67, 139, 34, 3, 37, 1, 13, 80, 10, 3, 248, 249, 50,
140, 250, 141, 251, 50, 140, 67, 14, 1, 67, 139, 58, 137,
23, 127, 128, 129, 20, 3, 23, 46, 38, 4, 26, 1, 25,
18, 1, 40, 11, 130, 3, 28, 8, 4, 63, 64, 59, 131,
5, 61, 65, 84, 33, 252, 60, 20, 41, 52, 119, 132, 7,
1, 253, 4, 47, 1, 142, 56, 110, 255, 8, 4, 24, 1,

```

```
y = padded_input_sequences[:, -1]
```

```
X.shape
```

```
(863, 56)
```

```
y.shape
```

```
(863,)
```

```
len(tokenizer.word_index)
```

```
282
```

```

from ast import Num
from tensorflow.keras.utils import to_categorical
y = to_categorical(y, num_classes=283)

```

```
y.shape
```

```
(863, 283)
```

```
y
```

```

array([[0., 1., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       ...,
       [0., 0., 0., ..., 1., 0., 0.],
       [0., 0., 0., ..., 0., 1., 0.],
       [0., 0., 0., ..., 0., 0., 1.]])

```

```

#building LSTM model architecture
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense

```

```

model = Sequential()
model.add(Embedding(283, 100, input_length= 56))
model.add(LSTM(150))
model.add(Dense(283, activation= 'softmax'))

```

```
model.compile(loss = 'categorical_crossentropy', optimizer='adam', metrics=['accuracy'])
```

```
model.summary()
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
embedding_2 (Embedding)	?	0 (unbuilt)
lstm_2 (LSTM)	?	0 (unbuilt)
dense_2 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

```
model.fit(X,y,epochs=100)
```

```
Epoch 1/100
27/27 ----- 5s 92ms/step - accuracy: 0.0660 - loss: 5.4367
Epoch 2/100
27/27 ----- 3s 92ms/step - accuracy: 0.0776 - loss: 5.0961
Epoch 3/100
27/27 ----- 2s 90ms/step - accuracy: 0.0776 - loss: 5.0105
Epoch 4/100
27/27 ----- 4s 136ms/step - accuracy: 0.0776 - loss: 4.9755
Epoch 5/100
27/27 ----- 3s 103ms/step - accuracy: 0.0776 - loss: 4.9086
Epoch 6/100
27/27 ----- 2s 90ms/step - accuracy: 0.0823 - loss: 4.7934
Epoch 7/100
27/27 ----- 3s 91ms/step - accuracy: 0.1205 - loss: 4.6232
Epoch 8/100
27/27 ----- 3s 90ms/step - accuracy: 0.1263 - loss: 4.4208
Epoch 9/100
27/27 ----- 4s 156ms/step - accuracy: 0.1657 - loss: 4.2096
Epoch 10/100
27/27 ----- 2s 90ms/step - accuracy: 0.1912 - loss: 4.0029
Epoch 11/100
27/27 ----- 2s 91ms/step - accuracy: 0.2317 - loss: 3.7955
Epoch 12/100
27/27 ----- 2s 90ms/step - accuracy: 0.2422 - loss: 3.5885
Epoch 13/100
27/27 ----- 3s 105ms/step - accuracy: 0.2862 - loss: 3.3796
Epoch 14/100
27/27 ----- 4s 134ms/step - accuracy: 0.3152 - loss: 3.1824
Epoch 15/100
27/27 ----- 2s 89ms/step - accuracy: 0.3534 - loss: 2.9986
Epoch 16/100
27/27 ----- 2s 90ms/step - accuracy: 0.3824 - loss: 2.7981
Epoch 17/100
27/27 ----- 2s 90ms/step - accuracy: 0.4299 - loss: 2.6264
Epoch 18/100
27/27 ----- 3s 127ms/step - accuracy: 0.4635 - loss: 2.4653
Epoch 19/100
27/27 ----- 3s 111ms/step - accuracy: 0.5017 - loss: 2.3114
Epoch 20/100
27/27 ----- 2s 89ms/step - accuracy: 0.5388 - loss: 2.1598
Epoch 21/100
27/27 ----- 3s 91ms/step - accuracy: 0.5713 - loss: 2.0219
Epoch 22/100
27/27 ----- 2s 91ms/step - accuracy: 0.6176 - loss: 1.8897
Epoch 23/100
27/27 ----- 4s 151ms/step - accuracy: 0.6257 - loss: 1.7747
Epoch 24/100
27/27 ----- 2s 90ms/step - accuracy: 0.6709 - loss: 1.6562
Epoch 25/100
27/27 ----- 3s 90ms/step - accuracy: 0.6871 - loss: 1.5479
Epoch 26/100
27/27 ----- 3s 90ms/step - accuracy: 0.7196 - loss: 1.4494
Epoch 27/100
27/27 ----- 2s 92ms/step - accuracy: 0.7428 - loss: 1.3569
Epoch 28/100
27/27 ----- 4s 150ms/step - accuracy: 0.7729 - loss: 1.2715
Epoch 29/100
27/27 ----- 2s 91ms/step - accuracy: 0.7903 - loss: 1.1932
```

```
text = 'mail'

for i in range(5):

    #steps
    #1.tokenize
```



```
'session': 34,
'our': 35,
'paid': 36,
'join': 37,
'able': 38,
'your': 39,
'website': 40,
'placement': 41,
'fee': 42,
'data': 43,
'monthly': 44,
'month': 45,
'not': 46,
'get': 47,
'yes': 48,
'once': 49,
'past': 50,
'feb': 51,
'assistance': 52,
'science': 53,
'7': 54,
'5600': 55,
'are': 56,
'watch': 57,
```

```
import time
text = 'nlp and deep learning'

for i in range(9):

    #steps
    #1.tokenize
    token_text = tokenizer.texts_to_sequences([text])[0]
    #padd
    padded_token_text = pad_sequences([token_text], maxlen = 56, padding = "pre")
    print(padded_token_text)
    #prediction
    model.predict(padded_token_text).shape
    pos = np.argmax(model.predict(padded_token_text))

    for word, index in tokenizer.word_index.items():
        if index == pos:
            text += " " + word
            print(text)
            time.sleep(2)
```

```
[[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 102 20
101 74]]
1/1 ██████████ 0s 51ms/step
1/1 ██████████ 0s 51ms/step
nlp and deep learning both
[[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 102 20 101
74 167]]
1/1 ██████████ 0s 49ms/step
1/1 ██████████ 0s 51ms/step
nlp and deep learning both are
[[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 102 20 101 74
167 56]]
1/1 ██████████ 0s 76ms/step
1/1 ██████████ 0s 72ms/step
nlp and deep learning both are not
[[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 102 20 101 74 167
56 46]]
1/1 ██████████ 0s 47ms/step
1/1 ██████████ 0s 46ms/step
nlp and deep learning both are not a
[[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 102 20 101 74 167 56
46 5]]
1/1 ██████████ 0s 50ms/step
1/1 ██████████ 0s 45ms/step
nlp and deep learning both are not a part
[[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
```

```

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 102 20 101 74 167 56 46
5 78]]
1/1 _____ 0s 46ms/step
1/1 _____ 0s 47ms/step
nlp and deep learning both are not a part of
[[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
   0 0 0 0 0 0 0 0 0 0 102 20 101 74 167 56 46 5
78 6]]
1/1 _____ 0s 51ms/step
1/1 _____ 0s 50ms/step
nlp and deep learning both are not a part of this

```